

Beijing-Dublin International College



AUTUMN TRIMESTER FINAL EXAMINATION - (2025/2026)

School of Computer Science

COMP3038J Multi-Paradigm Programming

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Time Allowed: 60 minutes

Instructions for Candidates:

Answer **All** questions.

BJUT Student ID:_____ **UCD Student ID:**_____

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge:_____ **(Signature)**

Instructions for Invigilators

Non-programmable calculators are permitted.
No rough-work paper is to be provided for candidates.

Question 1:

a. Explain the difference between the keywords `var` and `val`. Which of these should be preferred when programming in a functional style? Give an example of the use of both keywords including specifying the type of the variable being defined.

(10%)

b. Scala allows the definition of functions/methods with multiple parameter lists. Give an example of how you would call a function named `foo` with two parameter lists, both of which require a single int parameter. Give two benefits of using multiple parameter lists.
(10%)

c. Given variables called `name` and `price`, write an interpolated string expression to produce the following: `"Coke Zero $ 3.40"`.

Note the name should be left aligned in a space 20 characters wide, followed by the dollar sign, followed by the price right aligned in a space 10 characters wide with 2 digits after the decimal point. **(10%)**

d. Generic type parameters in Scala can be *invariant*, *covariant*, or *contravariant*, explain the difference between *invariant* and *covariant*. What is required in order to specify a covariant type parameter? Give an example of a generic type definition for *covariant* and *contravariant* type parameters. **(10%)**

e. Explain in detail what it means for a function to be considered **pure**. What are the benefits of using pure functions in our code? **(10%)**

(Question Total 50%)

Question 2:

a. Explain the benefits of using tail recursion. What is required in order for a function to be tail recursive?

Redefine the following function so that it is tail recursive.

```
1  def recFun(n: Int): Int =
2    if n == 1 then
3      5
4    else
5      recFun(n - 1) + 3
```

(10%)

b. Referential transparency is a property of expressions. Explain what this means. Include an example in your code of one expression that is referentially transparent and one that is not. **(10%)**

c. 1) Given a list of `Int` values named `numbers`, write an expression that will return a list containing only the numbers from the list that are divisible by 5.

2) Given a list of `Double` values named `floats`, write an expression that will return a list containing these numbers where each has been divided by 3.
3) Given a list of `Int` values named `nums`, write an expression that will return a list containing only the numbers less than 50, however these numbers should be modified such that each has been multiplied by 3.
4) Given a list of `Int` values named `vals`, write an expression that will return a list containing these values multiplied by 2, but only where the result of the multiplication is greater than 500.
5) Given a list of `String` values called `names`, write an expression that will return a list of integer values representing the length of the strings in the list.

(10%)

d. Currying is a mathematical technique for translating functions that take multiple parameters into a collection of families of functions that that each take a single parameter.

Show two examples of how to define a function that when given a single `Int` (a) parameter will return a function that takes an `Int` parameter and returns the value multiplied by a. The first example should be implemented using a closure, the second example should be implemented using multiple parameter lists.

(10%)

e. Data modelling is a core aspect of programming, create a "skinny domain model" for the messaging application described below.

This basic modelling system contains user accounts, direct messages, posts and comments. For each of these, the system remembers the following attributes:

- **User account** - username, email, and password.
- **Direct message** - message text, who it is to, who it is from and the date and time it was sent.
- **Post** - post text, who posted it, and the date and time it was posted.
- **Comment** - comment text, which post it was made on, who the comment was by, and the date and time it was posted.

For representing date and time you can assume you have the use of the class `DateTime`.

(10%)

(Question Total 50%)